

[DC] Embodying an avatar with an asymmetrical lower body to modulate the dynamic characteristics of gait initiation

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ABSTRACT

Virtual reality (VR) enables the user to perceive body ownership towards a virtual body. This illusion is induced through first-person perspective (1PP) and synchronous movement with the real body. Previous studies have shown that pronounced differences between the real and the virtual body lead to changes in the user's behavior. It has also been shown that modifying the body image can affect the user's movements. Nevertheless, the state of the art does not refer to the kinetic and kinematic impacts of one virtual lower limb deformation. Therefore, this paper presents a methodology exploring the impact of a self-avatar with an asymmetrical lower body (one limb longer or larger than the other) on the dynamic characteristics of the user during a gait initiation task.

Keywords: Self-avatar, virtual reality, ownership illusion, visual feedback, gait initiation, embodiment.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, Augmented, and Virtual Realities; H.5.2 [Information Interfaces and Presentation]: User Interfaces—Evaluation/Methodology.

1. INTRODUCTION

Virtual reality (VR) is an immersive experience where the user can perceive the generated environment as real. The user can then adapt his behavior according to events happening in the virtual world. In rehabilitation, the goal of motor adaptation is to reduce systematic error driven by a perturbation [1]. In VR, this perturbation can be represented as a change in the virtual environment (VE) or a change on the virtual body itself. Indeed, a user can embody an avatar that replicates his/her movements from a first-person perspective (1PP). Embodiment implies that the user feels that his/her own body is substituted by the body of the self-avatar.

Previous studies have shown that altering the morphology of one's self-avatar can lead to changes in behavior. Kilteni *et al.* [2] strongly altered the body representation of embodied participants by giving their self-avatar one very long arm. The ownership illusion was maintained towards this asymmetrical body and it led to a proprioceptive drift that was correlated to the length of the arm. Furthermore, modifying the body image can induce changes in one's body schema [3]. For example, embodying a giant monster modifies kinematics parameters during gait [4]. However, results from these previous studies do not focus on the motor consequences of a specific modification of a virtual limb.

The purpose of the present project is to radically modify an inertial parameter of one virtual lower limb during a gait initiation task. This asymmetry could generate dynamic compensations modifying the user's movements, demonstrating that an embodiment in a deformed avatar can alter motor behavior.

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2. CONTRIBUTION TO KNOWLEDGE

This project would be the first to investigate motor compensations as a result of embodying an asymmetrical avatar. It is a first step towards the establishment of a virtual training tool dedicated to rehabilitation, which would integrate a morphologically personalized self-avatar depending on the patient weaknesses. Indeed, the kinetic and kinematic modifications generated by the asymmetrical avatar could counterbalance motor deficits observed in a patient. The present project aims to verify the feasibility of this approach on healthy subjects and identify optimal parameters that could influence a user's movement.

3. GAIT INITIATION

The dynamic task used for this project is gait initiation. This task involves the integration of multisensory information including the visual, vestibular and proprioceptive systems. It is also a transition from standing to walking and involves a goal directed shift of the body [5]. Therefore, gait initiation is a demanding task frequently used in many studies to investigate how the central nervous system preserves the body balance when a perturbation appears. An adaptation to the asymmetrical self-avatar would then be translated to a modification in the controlling balance strategy.

4. OBJECTIVES

The first objective of this project is to evaluate the impact of different levels of embodiment of a self-avatar with one long or large tibia segment on the dynamic characteristics of the user. The second objective is to verify if the level of distortion of the tibia segment is correlated with the observed changes in the participants' kinetics and kinematics. Moreover, a measure of a proprioceptive drift of the foot will be tested to verify a possible correlation between lower-limbs motor and proprioceptive drifts. The third objective is to evaluate the impacts on kinetics and kinematics of the embodiment of an avatar with a below the knee prosthesis. Users may modify their movement because they perceive the prosthesis as heavier, demonstrating that motor modulations may not be strictly linked to a deformation of the self-avatar's morphology. Correlations with previous tibia distortions will also be evaluated.

5. FIRST STUDY METHODOLOGY

5.1. Participants

Based on previous studies of a similar nature we expect 40 participants will be sufficient to test our hypotheses.

5.2. Materials

Morphological features such as the height, weight, gender, skin color and the neck, bust and waist circumferences will be assessed to implement a personalized avatar using the MakeHuman software. The avatar will then be imported into a VE created in the Unity3D game engine. Participants will visualize a virtual replica of our research lab through a HTC Vive Pro head-mounted device (HMD) with a 1PP. The motion capture (MoCap) system will involve 12 optoelectronic cameras tracking 16 rigid marker clusters fitted on the participant (Figure 1A). Kinematic data thus obtained

will be captured by Vicon Tracker 3.6 and processed by Vicon Pegasus Advanced 1.2 to animate the avatar. Kinetic data will be recorded using a static AMTI split-belt treadmill. Post-analysis of the dynamic data will be done using Vicon Nexus 2.8.



Figure 1: **A:** participant wearing MoCap markers and observing his self-avatar's leg via the HMD; **B:** avatar with a long leg shown in 3PP; **C:** avatar with a large leg shown in 3PP; **D:** "normal" avatar stepping on the first target from the footprints.

5.3. Dependent variables

The biomechanical variables studied for this project are the position, speed, and acceleration of the center of mass (COM) and of the center of pressure (COP) from the stationary standing position to the first heel-strike. The subjective feeling of embodiment will be measured by a standardized questionnaire [6].

5.4. Proposed protocol

Participants will have to perform series of 2-step gait initiations with their preferred limb in 3 blocks of conditions: a normal self-avatar; a long tibial segment (Figure 1B); and a large tibial circumference (Figure 1C). In each block, gait will be initiated under two conditions of embodiment level: a high embodiment level created by synchronized visuotactile feedbacks before the trial, or a low embodiment level created by asynchronous visuotactile feedbacks. Twenty trials will be performed in each condition ($N = 200$ trials). In each condition, foot placement targets will be displayed either at 100% or at 80% of the participant's initial step length to eliminate variability between participants and conditions. The order of the blocks and conditions will be randomized across participants to avoid ranking effects.

In each condition, participants will first have to observe their self-avatar in a virtual mirror. Then they will have to stand in a comfortable position and their footprints will be displayed on the floor. These footprints will be used as a starting point for each condition, as the COM and COP during anticipatory postural adjustments (APA) highly rely on the initial standing position [5]. After the experimenter delivers an auditory cue, targets will appear on the virtual treadmill (Figure 1D) with a randomized delay (1 to 3 s) and participants will have to perform a 2-step gait initiation. A 6-minute rest will be imposed between two successive conditions to avoid fatigue. During this period, the participant will have to answer an embodiment questionnaire and walk in a corridor to eliminate any adaptation to the self-avatar.

5.5. Expected results

Our hypothesis is that the modification of the length of the virtual tibia will lead to the same modifications of the COM and the COP as adding weight to one leg during gait initiation [7]:

- Dynamic adjustments will focus on the mediolateral axis.
- To maintain balance, the COM will be transferred on the ipsilateral leg.
- COM and COP trajectories will have increased amplitudes during APA.

Moreover, modifying the tibial circumference will have the same dynamic effects as modifying the length of the tibia. Finally, strengthening the feeling of embodiment will lead to greater changes in the kinetic and kinematic data.

6. Conclusion

This paper presents a methodology, which enables the validation of motor modifications due to the embodiment of a self-avatar with an asymmetrical lower body. Gait initiation will allow to observe any modulation of the COM and COP positions, velocities and accelerations. The modulations would imply an adaptation to the asymmetrical body to counterbalance this perturbation.

The first study will ascertain whether any kinetic and kinematic changes in the gait initiation are correlated to the level of embodiment in the distorted avatar. Future studies will focus on the strength of this illusion and the possible correlation between motor and proprioceptive drifts. Furthermore, the same kinetic and kinematic analysis will be done for a self-avatar with a below the knee prosthesis. Some considerations would be valuable to discuss before undertaking further actions:

- Are the mentioned dependent variables sufficient to evaluate a possible motor adaptation?
- How can proprioceptive drift of the foot be reliably measured?
- Will the participants have a similar motor adaptation towards a below the knee prosthesis?

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